

Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor

10,194 shipments analysed	4.29 d avg processing lead time	14.0% delay frequency (SLA breach)	196 factory → state routes
-------------------------------------	---	--	--------------------------------------

Prepared by: Data Analyst Intern

Date: 02 October 2026

Data: 10,194 order lines, January 2024 – December 2025, 5 factories, 15 products, US & Canada

Tools: Python (pandas, SciPy), Plotly, Streamlit, Matplotlib

1. Executive summary

Nassau Candy ships confectionery from five factories to customers across the United States and Canada. This study measures how efficiently each factory → customer route performs, finds where delays concentrate, compares ship modes, and tests whether re-assigning production could shorten the distances products travel.

Bottom line. Delays are created inside the fulfilment operation, not on the road: processing lead time is driven by ship mode and by a sharp 2025 deterioration (SLA breach 7.4% → 18.7%), while route distance has no effect on it ($p = 0.01$). Distance matters for freight cost – and there the network is poorly aligned: routing chocolate orders to the nearer of the two chocolate plants would cut chocolate unit-miles by 49%.

Key findings

- **Data quality:** the raw Ship Date – Order Date gap is 904–1,642 days. It is a year-level date shift; the remainder (0–8 days) matches ship-mode SLAs exactly and is used as the true processing lead time.
- **Ship mode dominates speed:** Same Day 0.4 d → Standard Class 5.3 d. Standard Class carries 60% of volume and the highest breach rate (17.1%); First Class misses its SLA 14.6% of the time.
- **Service is deteriorating:** delay frequency rose from 6.4% in H2-2024 to 20.3% in H2-2025, in every ship mode.
- **Route leaderboard:** 58 routes with ≥ 30 shipments were scored. Best: short regional lanes such as Wicked Choccy's → Wisconsin and Wicked Choccy's → Ohio. Worst: Wicked Choccy's → Washington, Wicked Choccy's → Minnesota and Lot's O' Nuts → Connecticut.
- **Geographic bottlenecks:** Minnesota, Missouri, Utah, Oklahoma, Indiana, Washington are significantly slower than average after controlling for ship mode.
- **Network misalignment:** Lot's O' Nuts (Arizona) handles 56% of shipments but ships the furthest (1,333 mi on average).

Recommendations

#	Recommendation	Evidence	Expected impact
1	Root-cause the 2025 dispatch slowdown (staffing, carrier pick-up windows, order cut-off times) and add fulfilment capacity.	SLA breach rose from 7.4% (2024) to 18.7% (2025), inside every ship mode, with a stable mode mix.	Bring breach rate back toward the 2024 level.
2	Dual-source Wonka Bars from the Arizona and Georgia plants, routing each order to the nearer plant.	Cuts chocolate unit-miles by 49% (56% if Secret Factory is added as a third source).	Lower freight cost and real transit time; removes most cross-country lanes.
3	Protect First Class with a dedicated pick lane or earlier cut-off.	14.6% of First Class shipments miss the 3-day SLA – premium customers are let down roughly one time in seven.	Higher premium-service reliability and retention.
4	Targeted review of bottleneck states (Minnesota, Missouri, Utah, Oklahoma, Indiana, Washington, Illinois).	Significantly slower than average after controlling for ship mode (95% CI above zero).	Remove local carrier / hub issues.
5	Fix the date-shift defect in the order/ship-date export and add freight cost and delivery date to the data.	Raw Ship – Order gaps of 904–1,642 days; margins look identical across ship modes.	Trustworthy KPIs and true cost-to-serve monitoring.

2. Background and problem statement

Nassau Candy Distributor manufactures and distributes confectionery through a network of five factories. Each product is produced at exactly one factory and shipped to retail customers by one of four ship modes (Same Day, First Class, Second Class, Standard Class). Management had no route-level view of logistics performance: it could not say which factory → customer lanes are fast and reliable, which are slow, where geographic bottlenecks sit, or whether the factory network is well placed relative to demand.

Objectives

- Measure shipping lead time and delay frequency at route level (factory → state and factory → region).
- Rank routes on a single, fair efficiency score and identify the best and worst performers.
- Locate geographic bottlenecks with statistical confidence rather than raw averages.
- Compare ship modes on speed, reliability and margin.
- Quantify how much shipping distance could be saved by re-assigning production between factories.
- Deliver an interactive Streamlit dashboard so operations teams can explore the results themselves.

3. Data and preparation

The dataset contains 10,194 order lines with 18 fields: order and ship dates, ship mode, customer location (country, city, state/province, postal code, region), product and division, and sales, units, cost and gross profit. Two reference tables from the project brief were added: factory coordinates and the product → factory mapping (Appendix A).

Check	Result	Action
Missing values / duplicate rows	0 / 0	None needed
Sales – Cost = Gross Profit	Holds on every row (max error $\approx 1e-14$)	Financial fields trusted
US ZIP codes with 4 digits	449 rows (New England / New Jersey lost a leading zero)	Left-padded to 5 digits
Product → factory mapping	15 of 15 products mapped	Factory + coordinates joined
Customer geocoding	100% (US: ZIP centroid; Canada: city centroid)	Haversine distance computed
Ship Date – Order Date	904 – 1,642 days	Date-shift corrected (Section 3.1)

3.1 Lead-time data-quality finding

Computed literally, the gap between ship and order date is 2.5–4.5 years – impossible for confectionery. The gaps form three narrow bands starting at 904, 1,269 and 1,634 days – exactly 365 days apart. Inside each band, the leftover days follow the ship-mode hierarchy precisely: Same Day 0–2, First Class 1–4, Second Class 2–6, Standard Class 3–8. Such a pattern could not survive if the whole gap were noise, so the remainder is the true order-processing time and the band offset is a date shift introduced when the data was exported.

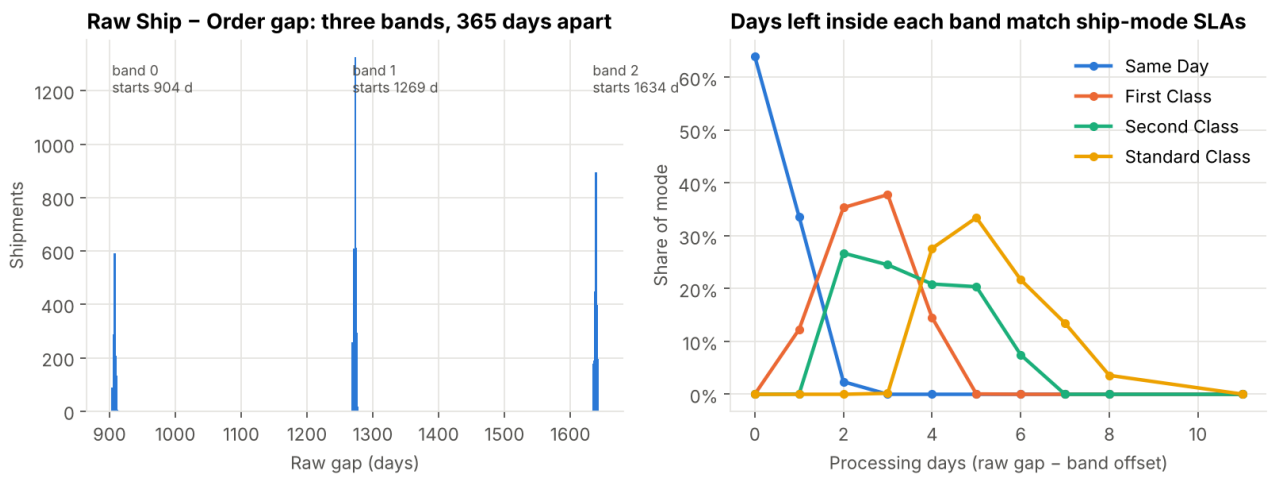


Figure 1. Left: raw gap in three 365-day bands. Right: processing days left inside each band, by ship mode - each mode sits in its expected range.

Decision. All KPIs use $lead_time_days = (raw\ gap - 904) \bmod 365$. The raw value is retained in the clean dataset and shown in the dashboard's Data & Method tab for full transparency.

4. Methodology and KPI definitions

KPI	Definition
Shipping lead time	Processing days from order to dispatch (date shift removed)
Average lead time	Mean lead time for a route, state, region or ship mode
Route volume	Number of shipments (order lines) on a route
Delay frequency	Share of shipments whose lead time exceeds the ship-mode SLA: Same Day ≤ 1 d, First ≤ 3 d, Second ≤ 5 d, Standard ≤ 6 d (≈ 75 th percentile of each mode). The dashboard also allows a single global threshold.
Excess days	Lead time minus the network average for the same ship mode – removes ship-mode mix so routes with many Standard Class orders are not unfairly penalised
Route efficiency score	0–100 (higher is better): 50% excess days, 30% delay frequency, 20% average distance, each percentile-ranked across routes with ≥ 30 shipments, then rescaled so the best route = 100 and the worst = 0
Distance	Great-circle (haversine) miles from factory to customer location

Analytical steps

- Clean and enrich the data (Section 3) in a reusable pipeline (*src/data_prep.py*).
- Aggregate KPIs by ship mode, month, factory $\rightarrow$ state route, factory $\rightarrow$ region route and state (*src/metrics.py*).
- Rank routes on the efficiency score with a minimum-volume rule (≥ 30 shipments) to avoid small-sample noise.
- Test geographic effects statistically: Kruskal-Wallis on excess days by region, factory and state; 95% confidence intervals for each state; Spearman correlation between distance and excess days.
- Simulate production re-assignment: unit-weighted distance if each product were made at each factory, and a nearest-plant fulfilment scenario for chocolate.
- Publish everything in a Streamlit dashboard with date, region, state, ship-mode, factory and delay-threshold filters.

5. Results

5.1 Ship-mode performance

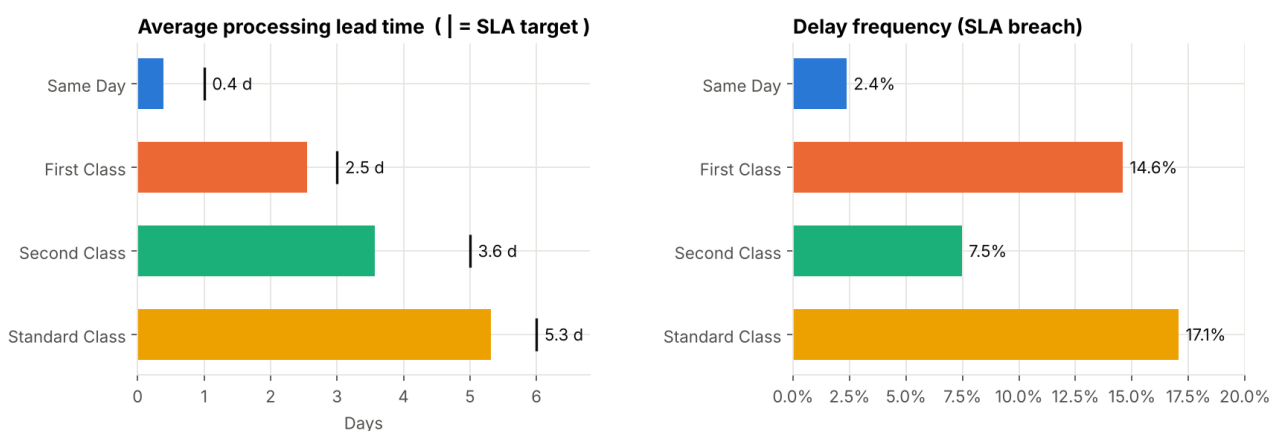


Figure 2. Average processing lead time against SLA target, and SLA breach rate, by ship mode.

Ship mode	Share	Avg lead (d)	SLA (d)	Delay rate	Gross margin
Same Day	5.4%	0.38	1	2.4%	66.1%
First Class	15.2%	2.55	3	14.6%	65.7%
Second Class	19.4%	3.57	5	7.5%	65.7%
Standard Class	60.0%	5.32	6	17.1%	66.0%

Ship mode is the dominant driver of lead time. Standard Class is both the largest and the least reliable mode. First Class is a concern: customers pay for speed yet one shipment in seven misses its 3-day target. Gross margin is flat across modes because freight cost is not recorded – faster modes look free in this data, which they are not.

5.2 Delay trend

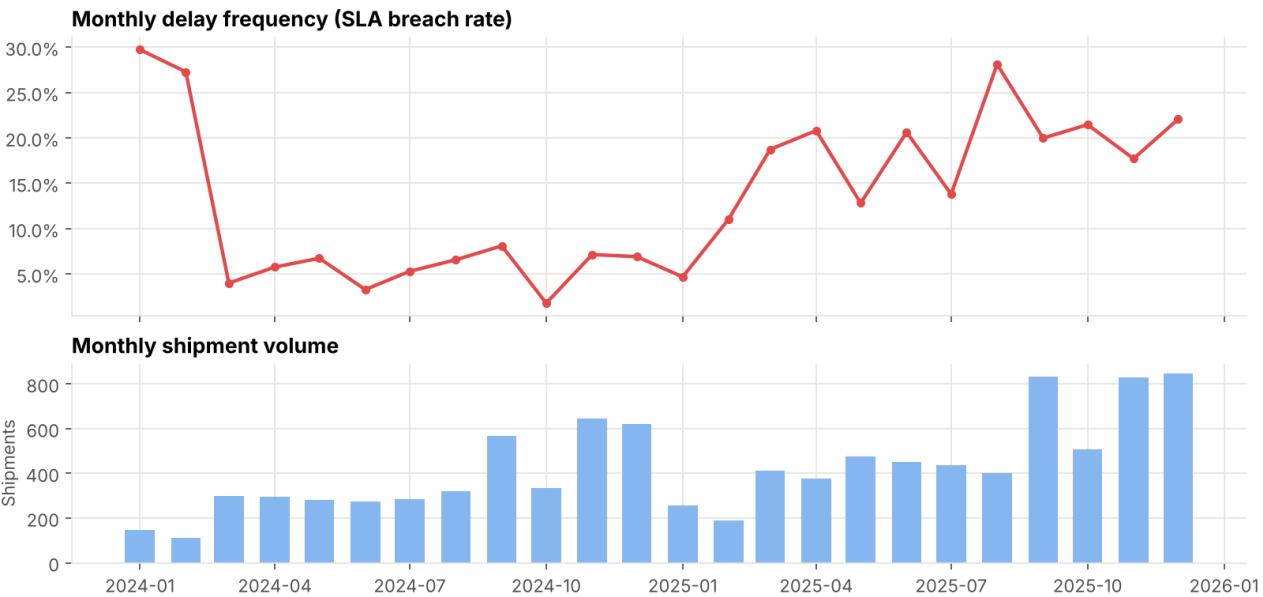


Figure 3. Monthly SLA breach rate (top) and shipment volume (bottom).

Delay frequency was low through 2024 ($\approx$ 5-8% after a small-sample January-February) and stepped up through 2025, ending the year at around 20%. The rise appears inside every ship mode while the ship-mode mix stayed nearly constant, so it is not caused by customers choosing slower modes. Monthly volume is only weakly related to the delay rate, pointing to a structural change in the fulfilment process in 2025 rather than seasonal peaks alone.

5.3 Route efficiency leaderboard

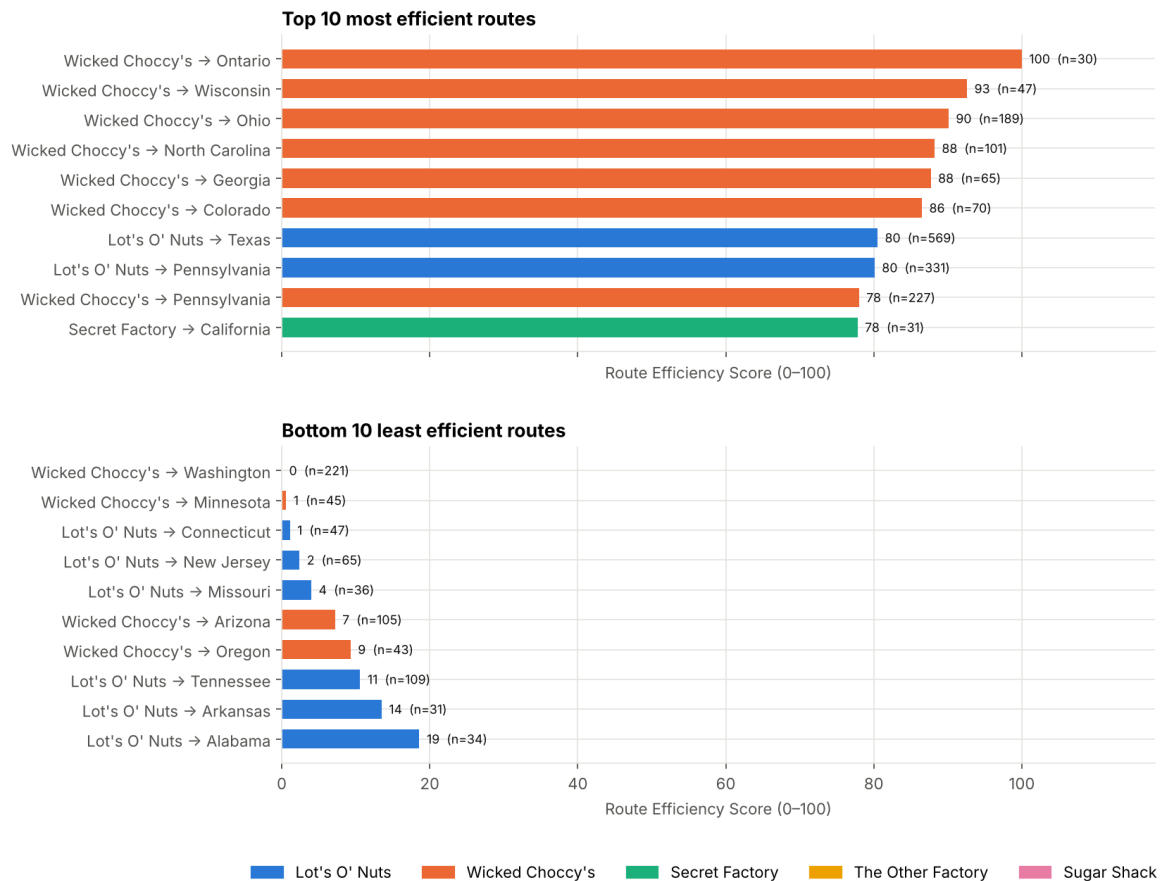


Figure 4. Top and bottom 10 of 58 scored factory → state routes (≥ 30 shipments). Bar colour = factory.

Route	Shipments	Avg lead (d)	Delay rate	Distance (mi)	Score
Wicked Choccy's → Ontario	30	3.90	0.0%	805	100
Wicked Choccy's → Wisconsin	47	4.40	8.5%	917	93
Wicked Choccy's → Ohio	189	3.80	9.5%	590	90
Wicked Choccy's → North Carolina	101	4.26	8.9%	259	88
Wicked Choccy's → Georgia	65	4.15	7.7%	222	88
Lot's O' Nuts → Missouri	36	4.58	22.2%	1,135	4
Lot's O' Nuts → New Jersey	65	4.89	20.0%	2,125	2
Lot's O' Nuts → Connecticut	47	4.00	23.4%	2,200	1
Wicked Choccy's → Minnesota	45	4.91	35.6%	1,091	1
Wicked Choccy's → Washington	221	4.33	20.8%	2,394	0

The best routes are short regional lanes with on-time dispatch – Wicked Choccy's (Georgia) into the Southeast, Ohio and Wisconsin, and Lot's O' Nuts (Arizona) into Texas and California. The weakest routes combine long distance with above-average delays: Wicked Choccy's into Washington, Oregon and Arizona; Lot's O' Nuts into New Jersey and Connecticut; and Interior lanes such as Lot's O' Nuts into Missouri, Tennessee and Arkansas. Wicked Choccy's → Minnesota has the worst reliability of any route. A striking pattern is that each of the two big chocolate plants serves the other plant's natural territory – the root cause addressed in Section 5.6.

5.4 Geographic bottlenecks

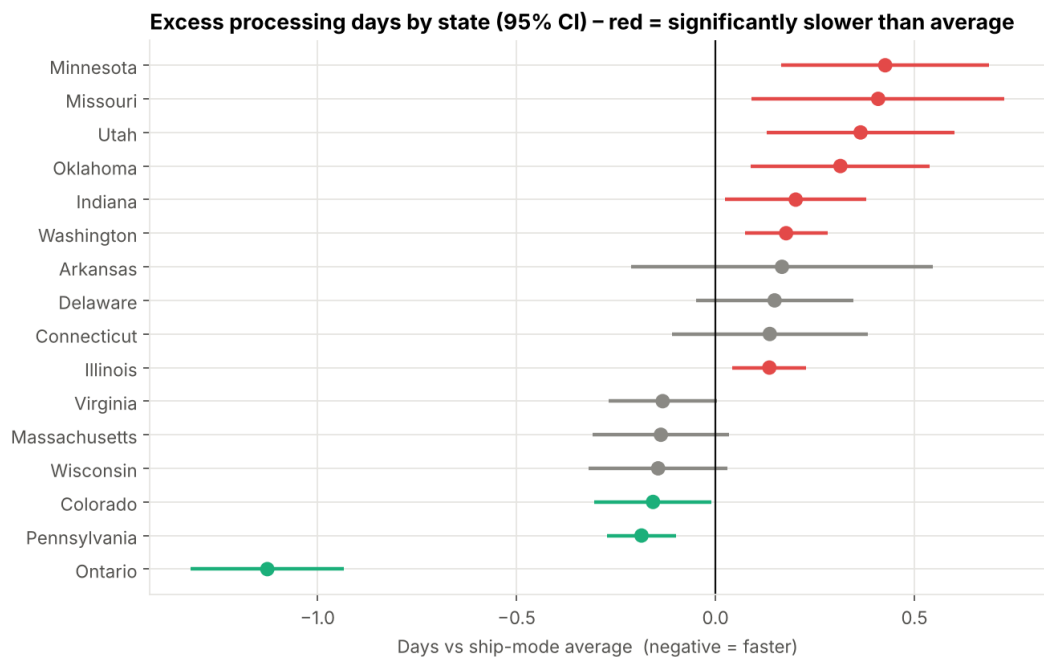


Figure 5. Excess processing days by state with 95% confidence intervals (states with ≥ 50 shipments; ten slowest and six fastest shown). Red = significantly slower than average.

State	Region	Shipments	Excess days	95% CI	Delay rate
Minnesota	Interior	89	+0.43	[+0.17, +0.69]	29.2%
Missouri	Interior	66	+0.41	[+0.09, +0.73]	16.7%
Utah	Pacific	53	+0.36	[+0.13, +0.60]	0.0%
Oklahoma	Interior	66	+0.31	[+0.09, +0.54]	12.1%
Indiana	Interior	149	+0.20	[+0.02, +0.38]	16.8%
Washington	Pacific	506	+0.18	[+0.07, +0.28]	18.6%
Illinois	Interior	492	+0.14	[+0.04, +0.23]	14.0%
New York	Atlantic	1128	+0.09	[+0.02, +0.15]	16.3%

A cluster of Interior and Upper-Midwest states – Minnesota, Missouri, Oklahoma, Indiana and Illinois – is significantly slower than the network, together with Utah and Washington in the West. Arkansas and Connecticut show high breach rates but on small samples whose confidence intervals cross zero, so they are flagged for monitoring rather than immediate action. Canadian provinces (Ontario in particular) perform well.

5.5 Does distance drive lead time?

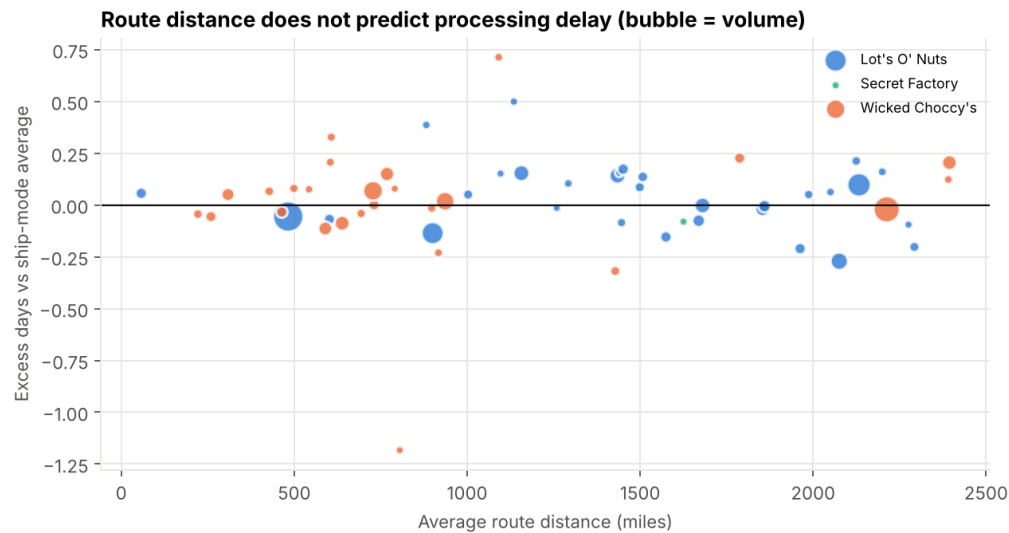


Figure 6. Route distance against mode-adjusted excess days; bubble size = shipments.

Test	Statistic	p-value	Conclusion
Spearman: distance vs excess days	$\rho = 0.007$	0.50	No relationship
Kruskal-Wallis by region	$H = 6.4$	9.2e-02	Not significant
Kruskal-Wallis by factory	$H = 5.1$	2.7e-01	Not significant
Kruskal-Wallis by state	$H = 193.2$	2.7e-22	Significant (small effect)

Distance has no relationship with processing lead time, and neither region nor factory differs significantly once ship mode is controlled for. State-level differences are statistically significant but small (within ± 0.5 days). The measured lead time is therefore dispatch time, not transit time: delays originate inside the fulfilment operation. This separates the problem into two levers – **process** (speed and reliability) and **network** (distance, freight cost and real transit time).

5.6 Factory network and sourcing scenarios

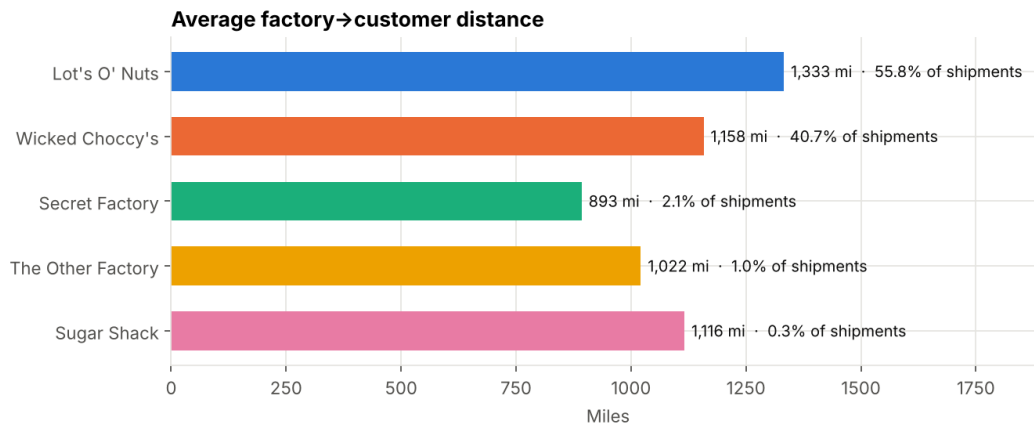


Figure 7. Average factory → customer distance and share of shipments.

Lot's O' Nuts in Arizona produces three of the five Wonka Bars and handles most shipments, yet ships the furthest – a large share of its customers are on the East Coast. Wicked Choccy's in Georgia, in turn, ships heavily to California and Washington. Two scenarios were tested:

- **Single-factory relocation** (Appendix B): moving any one Wonka Bar line entirely to the central Secret Factory (Illinois) would cut its unit-miles by 18–30%, but would overload a factory that currently makes ~2% of volume.
- **Nearest-plant fulfilment:** each chocolate order is made at whichever of Lot's O' Nuts or Wicked Choccy's is closer. Chocolate unit-miles fall from 47.1 M to 24.0 M (–49%); 52% of orders would switch plant. Adding Secret Factory as a third source raises the saving to 56%.

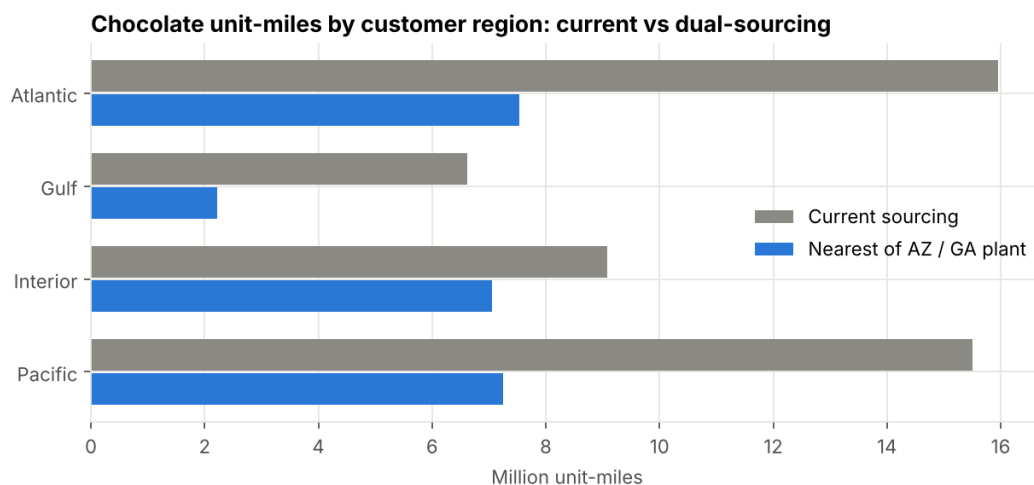


Figure 8. Chocolate unit-miles by customer region: current sourcing vs nearest-plant fulfilment.

6. Recommendations

#	Recommendation	Evidence	Expected impact
1	Root-cause the 2025 dispatch slowdown (staffing, carrier pick-up windows, order cut-off times) and add fulfilment capacity.	SLA breach rose from 7.4% (2024) to 18.7% (2025), inside every ship mode, with a stable mode mix.	Bring breach rate back toward the 2024 level.
2	Dual-source Wonka Bars from the Arizona and Georgia plants, routing each order to the nearer plant.	Cuts chocolate unit-miles by 49% (56% if Secret Factory is added as a third source).	Lower freight cost and real transit time; removes most cross-country lanes.
3	Protect First Class with a dedicated pick lane or earlier cut-off.	14.6% of First Class shipments miss the 3-day SLA – premium customers are let down roughly one time in seven.	Higher premium-service reliability and retention.
4	Targeted review of bottleneck states (Minnesota, Missouri, Utah, Oklahoma, Indiana, Washington, Illinois).	Significantly slower than average after controlling for ship mode (95% CI above zero).	Remove local carrier / hub issues.
5	Fix the date-shift defect in the order/ship-date export and add freight cost and delivery date to the data.	Raw Ship – Order gaps of 904–1,642 days; margins look identical across ship modes.	Trustworthy KPIs and true cost-to-serve monitoring.

Suggested sequencing

- **0–30 days:** correct the date export (Rec. 5); set up weekly monitoring of breach rate by ship mode and the bottleneck states using the dashboard.
- **30–90 days:** root-cause the 2025 slowdown and pilot a First Class pick lane (Recs. 1 and 3).
- **3–9 months:** pilot nearest-plant fulfilment for one Wonka Bar line, then extend after confirming capacity and freight-cost savings (Rec. 2).

7. Streamlit dashboard

An interactive dashboard (*app.py*) exposes every KPI in this report. Global sidebar filters: order date range, region, state/province, ship mode, factory, delay definition (ship-mode SLA or a single lead-time threshold slider) and the minimum route volume for ranking.

Tab	What it shows
Overview	Headline KPIs, monthly delay trend and volume, factory summary
Route Efficiency	Top/bottom route charts and a sortable scorecard (state or region grain), CSV export
Geographic Map	US choropleth (delay, lead time, excess days, volume, distance), factory locations, route lines, regional bottleneck table and statistically slow states
Ship Modes	Lead time vs SLA, delay rate, lead-time distribution, ship mode × region heatmap
Route Drill-Down	Per-route KPIs, monthly trend, ship-mode mix, city table, order-level timeline, CSV export
Sourcing Scenarios	Interactive nearest-plant simulation and single-factory relocation table
Data & Method	Date-shift evidence and KPI definitions

8. Limitations and next steps

- Lead time measures order-to-dispatch, not delivery; adding a delivery date would allow true transit-time analysis.

- Distances are straight-line (great-circle); road distance is typically 15–25% longer but ranks routes the same way.
- Freight cost is absent, so ship-mode margins and scenario savings are expressed in miles, not dollars.
- The product → factory mapping and factory coordinates come from the project brief; factory capacity is not modelled.
- Sugar and Other divisions are small (350 shipments in total), so their route results are indicative only.

Appendix A - Reference data

Factory	Latitude	Longitude	State
Lot's O' Nuts	32.881893	-111.768036	Arizona
Wicked Choccy's	32.076176	-81.088371	Georgia
Sugar Shack	48.119140	-96.181150	Minnesota
Secret Factory	41.446333	-90.565487	Illinois
The Other Factory	35.117500	-89.971107	Tennessee

Product	Factory
Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts
Wonka Bar - Fudge Mallows	Lot's O' Nuts
Wonka Bar -Scrumdiddlyumptious	Lot's O' Nuts
Wonka Bar - Milk Chocolate	Wicked Choccy's
Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's
Laffy Taffy	Sugar Shack
SweeTARTS	Sugar Shack
Nerds	Sugar Shack
Fun Dip	Sugar Shack
Fizzy Lifting Drinks	Sugar Shack
Everlasting Gobstopper	Secret Factory
Lickable Wallpaper	Secret Factory
Wonka Gum	Secret Factory
Hair Toffee	The Other Factory
Kazookles	The Other Factory

Appendix B - Single-factory relocation check

Product	Units	Current factory	Avg mi	Closest factory	Avg mi	Saved
Wonka Bar -Scrumdiddlyumptious	7,743	Lot's O' Nuts	1,339	Secret Factory	951	29%
Wonka Bar - Nutty Crunch Surprise	6,755	Lot's O' Nuts	1,328	Secret Factory	931	30%
Wonka Bar - Fudge Mallows	6,914	Lot's O' Nuts	1,330	Secret Factory	950	29%
Wonka Bar - Triple Dazzle Caramel	7,596	Wicked Choccy's	1,186	Secret Factory	956	19%
Wonka Bar - Milk Chocolate	8,267	Wicked Choccy's	1,161	Secret Factory	952	18%
Laffy Taffy	27	Sugar Shack	1,273	Wicked Choccy's	591	54%
SweetARTS	41	Sugar Shack	1,074	Secret Factory	773	28%
Fizzy Lifting Drinks	21	Sugar Shack	1,131	Wicked Choccy's	746	34%
Wonka Gum	478	Secret Factory	803	The Other Factory	790	2%
Nerds	10	Sugar Shack	1,199	Wicked Choccy's	711	41%
Hair Toffee	17	The Other Factory	1,186	Lot's O' Nuts	970	18%
Fun Dip	8	Sugar Shack	959	Secret Factory	554	42%
Everlasting Gobstopper	13	Secret Factory	810	The Other Factory	639	21%
Lickable Wallpaper	393	Secret Factory	939	Secret Factory	939	0%
Kazookles	371	The Other Factory	988	The Other Factory	988	0%

Unit-weighted great-circle miles from each candidate factory to the product's actual customers.